CSS 2024 Student Paper Award: Fourier Spectrum Clone Detection in Nano-Artifacts
Threat of Physical Cloning & Counterfeit Hardware
Advanced 3D micro-printing, optical lithography, and reverse-molding threaten identity tokens, smart cards, and mission-critical hardware chips
Nano-Artifact Metrics (PUF)
• Naturally random metallic/dielectric particles;
• Microscopic surface variations during curing;
• Structurally impossible to duplicate (Unclonable)
The Challenge of Optical Noise
• Tilting, lighting jitter, and focus drift;
• Spatial domain pixel matching causes false rejects;
• High-speed verification required
YNU SPIRAL Innovation (CSS 2024 Student Paper Award: Miyamoto et al.)
Fourier Spectrum-Based Clone Detection for Nano-Artifact Metrics
Converts optical microscopy texture images into the 2D spatial frequency domain (Fourier Spectrum);
Isolates invariant physical structural patterns while filtering spatial translation and sensor noise
Zero-Trust Physical Authentication with Unconditional Uniqueness
Instantaneous discrimination between genuine nano-patterns and high-precision replicas;
Secures national defense passports, semiconductor supply chains, and ultra-secure hardware keys